

Approach

This project looks at how well Large Language Models (LLMs) can identify safety risks in autonomous driving corner-case scenarios. The main idea is to compare two approaches, a regular baseline method and an ontology-based method, to see if adding structured semantic knowledge improves how consistent, accurate, and reliable the risk detection is.

To do this, I made use of the batch-processing pipeline that works with sequences of images from driving scenarios. Each scenario contains multiple frames (like a short video), showing different conditions such as foggy roads, nighttime driving, or busy urban streets. These images were split into batches of different sizes (1, 3, and 5 frames) so I could also check how having more batches affects the model's performance.

For each batch, the model was asked to:

- identify important entities in the scene (like vehicles, pedestrians, and obstacles)
- describe how these entities relate to each other (for example, occlusion or proximity)
- generate a structured version of the scene using RDF triples in Turtle format
- list possible safety risks along with a severity score

A key part of this project is the use of ontology to guide the model's outputs. The ontology was designed to represent driving scenarios in a structured way, with clearly defined classes such as Vehicle, Pedestrian, RoadObject, TrafficSignal, WeatherCondition, and Risk. The Risk class was further expanded into more specific types like CollisionRisk, ReducedVisibilityRisk, LaneIntrusionRisk, PedestrianCollisionRisk, OcclusionRisk, and LowFrictionRisk.

The ontology also defines relationships between these concepts using properties such as:

- hasActor (linking scenes to vehicles or pedestrians)
- hasObstacle / hasRoadObject
- hasWeatherCondition
- hasRisk (linking the scene to identified risks)

By including this ontology directly in the prompt, the model is restricted to using only the allowed classes and relationships. This ensures that all outputs follow a consistent structure and vocabulary, instead of using free-form descriptions.

Two different prompting approaches were used:

- a baseline prompt, where the model had more freedom in how it described things.
- an ontology-based prompt, where the model had to follow predefined classes and relationships from the ontology.

The goal was to see whether the ontology-based approach helps reduce hallucination and produce more structured and consistent outputs, even if it slightly limits flexibility. Overall, this approach allows for a direct comparison between unstructured and ontology-guided reasoning in safety-critical scenarios

Ontology design

The ontology was developed to model autonomous driving corner-case scenarios, focusing on a structured representation of entities, environmental conditions, and safety risks. The design is directly inspired by the taxonomy proposed in the reference paper which classifies corner cases into three main categories: state anomalies, behavior anomalies, and evidence-based anomalies.

Based on this, the ontology captures semantic relationships between road actors, objects, and contextual factors that contribute to potential hazards

Key Classes

- Scene – Represents the overall driving environment
- Actor – Includes dynamic entities such as vehicles and pedestrians
- Vehicle and Pedestrian – Subclasses of Actor
- RoadObject – Includes static elements such as lanes, signs, and obstacles
- TrafficSignal and TrafficSign – Represent traffic control elements
- WeatherCondition – Captures environmental context (e.g., Rainy, Foggy, Night)
- EvidenceIndicator – Represents early signals of potential hazards (e.g., a ball on the road)
- Risk – Central class representing potential safety hazards

Corner Case Taxonomy Integration

A key design decision was to explicitly model the three anomaly types defined in the paper:

- State Anomaly - mismatch between object appearance and meaning
 - Example: advertisement resembling a STOP sign
- Behavior Anomaly - unusual or illegal behavior
 - Example: car driving in the wrong direction
- Evidence-Based Anomaly - early indicators of a potential hazard
 - Example: a ball indicating a child may enter the road

Risk Modeling

The ontology includes detailed subclasses of the Risk class to ensure consistent risk representation. Instead of using free-text descriptions, risks are modeled as explicit classes, which improves consistency and allows direct evaluation from RDF outputs. Some of the risk classes are

- CollisionRisk
- ReducedVisibilityRisk
- LaneIntrusionRisk
- PedestrianCollisionRisk
- OcclusionRisk
- LowFrictionRisk

Key Properties

The ontology defines relationships using object and datatype properties:

Object Properties:

- hasActor – links a scene to vehicles or pedestrians
- hasVehicle / hasPedestrian – more specific actor relationships
- hasRoadObject / hasObstacle – connects scene to static elements
- hasWeatherCondition – captures environmental context
- hasRisk – links a scene to identified risks

Datatype Properties:

- hasSeverityScore – numerical value (0–1) representing risk severity
- hasDescription – textual explanation of entities or risks

Design Choices

- A hierarchical structure was used (e.g., Actor → Vehicle, Pedestrian) to reflect real-world relationships.
- The ontology follows the paper’s anomaly taxonomy, ensuring alignment with the dataset and scenarios.
- Risks were modeled as classes instead of text, enabling structured evaluation from RDF.
- Environmental context (e.g., fog, night) was explicitly modeled to support scenario reasoning.
- Relationships were kept simple and interpretable, making it easier to parse RDF outputs.

These design choices were made to balance expressiveness and simplicity. Using a hierarchical structure allows ontology to capture real-world relationships more naturally, while aligning with the paper’s taxonomy ensures consistency with the scenarios being evaluated. Representing risks as explicit classes instead of free text helps standardize outputs and makes it easier to compute evaluation metrics directly from RDF. Additionally, keeping relationships simple improves interpretability and ensures that the generated output can be reliably parsed and analyzed.

Extensions Beyond the Paper

Extensions were introduced to make the ontology more expressive and better suited for real-world driving scenarios. While the original taxonomy provides a strong high-level classification, it does not fully capture the variety of risks encountered in complex environments. By adding finer-grained risk types and additional context classes, the ontology can represent scenarios more accurately. This also helps the model generate more meaningful and structured outputs, improving both semantic richness and evaluation quality.

- More detailed risk categories are used to capture finer safety risks that are not explicitly defined in the original taxonomy
 - SuddenObstacleRisk
 - UnpredictableBehaviorRisk
 - LowFrictionRisk

- Additional environmental context is added to better represent real-world driving conditions that directly affect perception and decision-making
 - NightCondition
 - FoggyCondition
- Evidence indicators were included to model early warning signals that may indicate the presence of hidden or upcoming risks
 - RollingBallIndicator
 - OccludedObjectIndicator

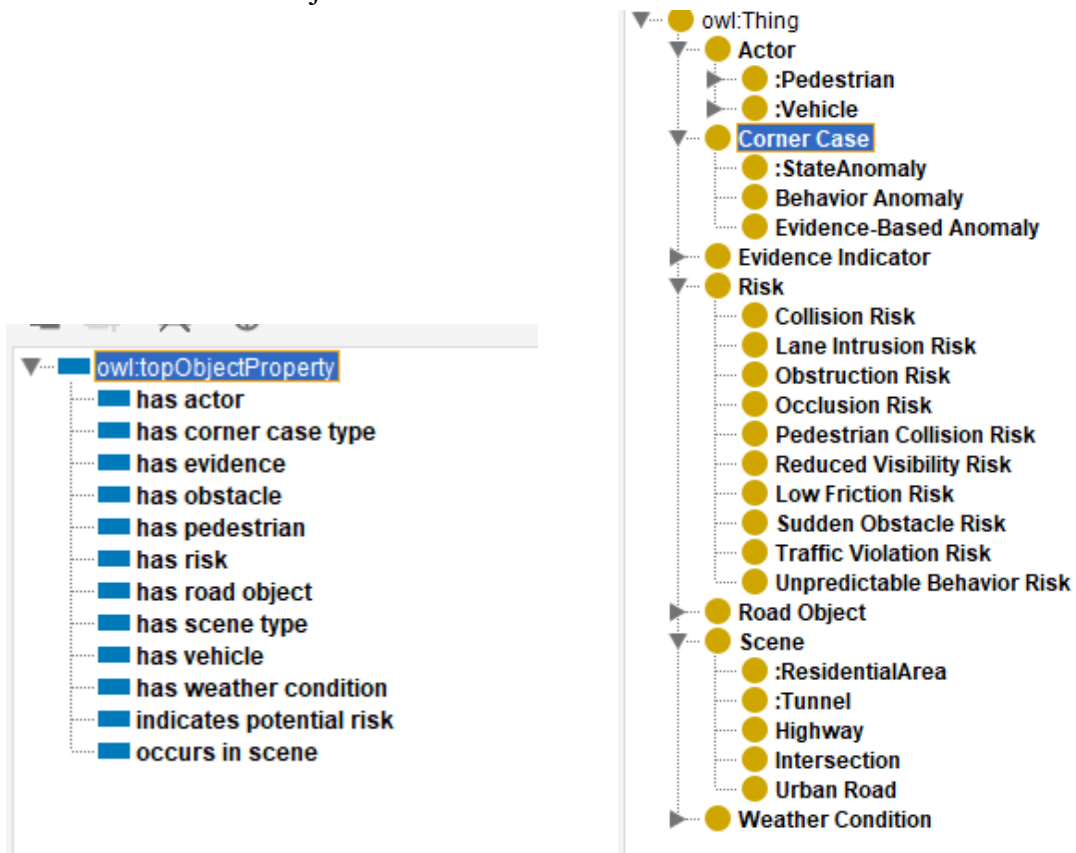


Figure 1: Ontology Class Hierarchy and Object Properties

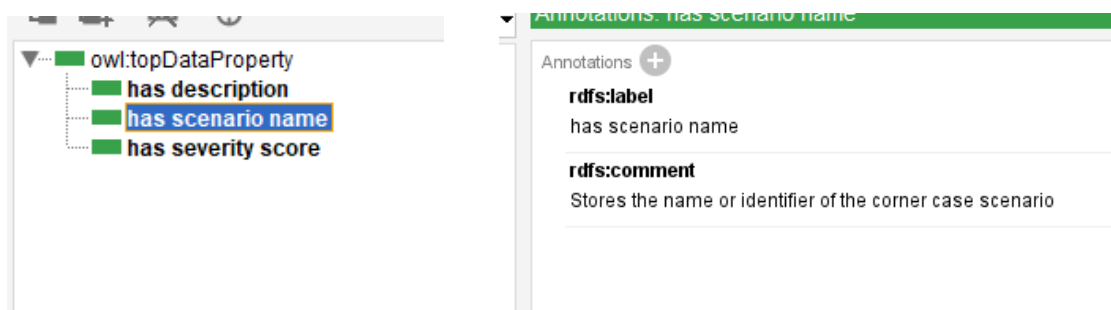


Figure 2 : Ontology Data Properties and label and comment

Parser Design

The parser was designed as a structured pipeline to transform both the ontology and the model-generated outputs into a format suitable for evaluation. It consists of two main components, an ontology parsing module and an output parsing module, with helper functions to organize the extracted information.

Ontology Parsing and Representation

The ontology parser is responsible for loading and structuring the ontology defined in Turtle (TTL) format. Using the `rdflib` library, the ontology is parsed into a graph representation, allowing access to all RDF triples. From this graph, key semantic elements are extracted, including classes, subclass relationships, object properties, and datatype properties.

Each class is stored along with its label and descriptive comment, enabling semantic interpretation beyond simple identifiers. Subclass relationships are extracted to preserve the hierarchical structure of the ontology, particularly for critical concepts such as risk types and evidence indicators. Similarly, object and datatype properties are captured along with their domain and range, ensuring that relationships between entities remain semantically valid.

Helper functions such as `short_name`, `get_label`, and `get_comment` are used to simplify URIs into readable identifiers and retrieve descriptive metadata. This step is essential for converting the ontology into a human-readable and model-friendly format.

Since the full ontology can be huge and difficult to directly incorporate into prompts, a summarization mechanism is implemented. The `summarized_ontology` function organizes the ontology into structured sections, including corner case categories, main scene concepts, risk concepts, evidence indicators, and object or road-related concepts. Each concept is optionally accompanied by its descriptive comment to provide semantic context.

In addition to the summary, the parser generates three supporting components: a list of allowed classes, a list of allowed relationships (object properties with domain and range), and detailed class descriptions derived from ontology comments. These components are used to constrain the language model during generation, reducing the likelihood of hallucinated classes or invalid relationships while promoting consistency with the ontology schema.

Output Parsing and Risk Extraction

The second component of the parser focuses on extracting structured information from the model-generated outputs. Since the system is designed to produce RDF triples in Turtle format, the primary extraction method operates directly on the TTL output. The RDF output is parsed using `rdflib`, and risk-related entities are identified by traversing triples associated with risk classes.

For each detected risk, the parser extracts the risk type, its natural language description, and the associated severity score. Duplicate risks are filtered to ensure that each risk is counted only once.

during evaluation. This structured extraction enables direct comparison with ground truth annotations for computing evaluation metrics such as precision, recall, and F1-score.

Fallback Mechanism

To improve robustness, a fallback parsing mechanism is implemented using regular expressions. In cases where the Turtle output generated is syntactically invalid or cannot be parsed correctly, the system reverts to extracting risks from the raw textual output. While this method is less semantically rigorous, it ensures that useful information can still be recovered and included in the evaluation process.

Prompt Strategy

In the ontology-based approach, the model was guided using structured information from the ontology instead of relying only on free-form reasoning. The prompt included a summarized schema, allowed classes, allowed relationships, and class descriptions, all generated from the ontology. This helped the model use consistent terminology and avoid random or incorrect concepts.

The model was asked to identify entities in the scene (such as vehicles, pedestrians, and obstacles), describe relationships between them, and classify the scenario into one of the three corner case types: StateAnomaly, BehaviorAnomaly, or EvidenceBasedAnomaly.

It was also required to generate RDF triples in Turtle format using ontology-defined classes and relationships. For each risk, the model had to follow a fixed structure including the risk type, description, and severity score. This made it easier to extract and evaluate risks later.

To improve reliability, the prompt included constraints to avoid hallucination. The model was instructed to only include risks supported by the images and to map any unclear risks to the closest ontology-defined class.

Overall, this approach helped produce more structured, consistent, and reliable outputs.

Prompt Used:

You are an autonomous driving safety analyst.

Your task is to analyze the following scene images and produce a semantic interpretation of the environment. The images represent a potential corner case scenario.

Use the ontology information below to guide your reasoning and ensure semantic consistency.

1. SUMMARIZED ONTOLOGY SCHEMA

{ontology_summary}

2. ALLOWED CLASSES

{allowed_classes}

3. ALLOWED RELATIONSHIPS

{allowed_relationships}

4. CLASS DESCRIPTIONS

{class_descriptions}

TASK

1. Identify the main entities in the scene:
(vehicles, pedestrians, cyclists, road infrastructure, obstacles, weather conditions)
 - Prefer ontology-defined classes
2. Identify relationships between entities:
(approaching, following, crossing, occluding, near)
 - Prefer ontology-defined relationships
3. Classify the scenario into ONE corner case type:
 - StateAnomaly
 - BehaviorAnomaly
 - EvidenceBasedAnomaly
4. Generate RDF triples describing the scene:
 - Use Turtle format
 - Ensure consistency with ontology classes and relationships
5. Identify potential safety risks:
 - Use ontology-defined risk concepts where possible
 - If the exact risk is not available, map to the closest ontology risk class
 - Describe the specific risk clearly based on what is visible in the images
 - Do NOT invent risks that are not supported by the images
 - Clearly NAME each risk
6. Assign a severity score between 0 and 1:
 - 0 = very low risk
 - 1 = very high risk

CRITICAL RDF REQUIREMENT

For EVERY risk in the Turtle output, you MUST use this exact structure:

```
:Risk_1 a :SomeRiskType ;  
  :hasDescription "Clear image-based description of the risk." ;  
  :hasSeverityScore "0.85"^^xsd:float .
```

Rules:

- Every risk MUST have:
 - rdf:type (or 'a') with a valid Risk class
 - hasDescription (MANDATORY)
 - hasSeverityScore (MANDATORY)
- Do NOT create risks without descriptions
- Each risk must be a separate RDF resource

IMPORTANT

- Use ontology concepts as guidance, not strict limitation
- Only include risks that are visually supported by the images

- Avoid hallucinated or unsupported entities, relationships, or risks
- Maintain consistency across entities, RDF triples, and risks

OUTPUT FORMAT

Scene Description in RDF Triples (Turtle Format)

Identified Risks

Risk 1:

Name:

Description:

Severity Score:

Risk 2:

Name:

Description:

Severity Score:

""""

Experimental Comparison

Item	Count
No. of Classes	79
No. of Object Properties	12
No. of datatype properties	3
No. of labels	54
No. of comments	54

Table 1.: Ontology Summary

Table 1 provides an overview of the key components of the developed ontology. Table 2 compares the performance of baseline and ontology-grounded approaches across several key aspects.

Scenario	Without Ontology	With Ontology	Improvement Observed
Semantic Consistency	Outputs are descriptive but vary across batches and use inconsistent terminology	Outputs follow predefined ontology classes and consistent structure	Improved consistency and standardized risk representation
Risk Accuracy	Higher or moderate F1 scores due to detecting more risks	Slightly lower F1 in some scenarios due to limited vocabulary	More precise and semantically aligned risk identification, though slightly less flexible
Hallucination Rate	Higher hallucination rates (e.g.: ~ 0.09 - 0.44 across models)	Reduced hallucination rates (e.g.: ~0.00 – 0.13 for most cases)	Significant reduction in unsupported or incorrect risks
RDF Quality	RDF outputs are less structured and may not follow a schema	RDF strictly adheres to ontology classes and relationships	Improved semantic structure correctness and interpretability

1. Semantic Consistency

The baseline approach produced descriptive outputs but often used inconsistent terminology across batches and scenarios. For example, similar risks were expressed using different names such as PerceptionErrorRisk, LaneAmbiguityRisk, and TrafficInteractionRisk, even when referring to conceptually similar hazards. In contrast, the ontology-grounded approach enforced the use of predefined classes such as CollisionRisk, OcclusionRisk, and ReducedVisibilityRisk, resulting in standardized and semantically consistent outputs. This consistency significantly improves interpretability and makes the outputs easier to compare across scenarios.

2. Risk Discovery Performance

Across models, the baseline approach generally achieved higher recall and sometimes higher F1 scores, as it identified a broader range of risks. However, this came at the cost of lower precision due to the inclusion of unsupported or loosely defined risks.

For example:

- In the FallenTree scenario (ChatGPT 5.4 mini), hallucination decreased from 0.44 (baseline) to 0.17 (ontology), but F1 also decreased from 0.69 to 0.57
- In StaticHighwayAd4 (Gemini), hallucination reduced from 0.09 to 0.04, while F1 slightly decreased from 0.87 to 0.79

This indicates that the baseline model tends to over-predict risks, increasing recall but introducing false positives. The ontology-grounded approach, on the other hand, produces more precise and semantically valid risk predictions, even if it occasionally misses less obvious risks.

3. Hallucination Reduction

One of the most significant improvements observed with ontology grounding is the reduction in hallucinated risks across all models.

- Gemini 3 Flash Preview shows the strongest improvement, with hallucination rates dropping to as low as 0.00–0.04 in several scenarios
- ChatGPT 5.4 mini shows a reduction from 0.44 to 0.17 in the FallenTree scenario
- LLaMA also shows improvement (e.g., 0.58 to 0.41), although its overall hallucination rate remains higher than the other models

This demonstrates that ontology grounding effectively constrains the model to generate only supported and valid risks, which is critical in safety-critical applications like autonomous driving.

4. RDF Output Quality

The baseline approach often generated RDF triples that were:

- inconsistent in structure

- loosely aligned with semantic relationships
- occasionally missing required properties

In contrast, the ontology-grounded approach produced RDF outputs that:

- strictly adhered to predefined classes and relationships
- consistently included required properties such as hasDescription and hasSeverityScore
- followed a standardized schema

This resulted in cleaner, more structured RDF outputs, which were significantly easier to parse and evaluate programmatically.

5. Model-Specific Observations

- Gemini 3 Flash Preview: Achieved the best balance between accuracy and reliability. It maintained competitive F1 scores while significantly reducing hallucination, making it the most robust model overall.
- ChatGPT 5.4 mini: Demonstrated strong baseline performance but experienced a noticeable drop in F1 when constrained by the ontology. However, hallucination reduction was still substantial.
- LLaMA: Showed the largest performance drop under ontology constraints, with lower F1 scores across all scenarios. While hallucination was reduced, the model struggled with constrained reasoning compared to the others.

Table 2.: Baseline vs Ontology-Grounded Comparison

Gemini			
Scenario Name	Ground Truth Risk	Baseline Output	Ontology Grounded Output
StaticHighwayAd4 NightRainy StateAnomaly	ReducedVisibilityRisk LowFrictionRisk LaneIntrusionRisk TrafficViolationRisk	ReducedVisibilityRisk LowFrictionRisk PerceptionErrorRisk TrafficViolationRisk	ReducedVisibilityRisk LowFrictionRisk CollisionRisk TrafficViolationRisk
FallenTree DayClear EvidenceBasedAnomaly	ObstructionRisk OcclusionRisk TrafficViolationRisk CollisionRisk	ObstructionRisk OcclusionRisk CollisionRisk LaneIntrusionRisk	ObstructionRisk CollisionRisk OcclusionRisk TrafficViolationRisk
FootballFollowedByBoy DayFoggy EvidenceBasedAnomaly	ReducedVisibilityRisk OcclusionRisk SuddenObstacleRisk PedestrianCollisionRisk UnpredictableBehaviorRisk	ReducedVisibilityRisk SuddenObstacleRisk PedestrianCollisionRisk CollisionRisk	PedestrianCollisionRisk ReducedVisibilityRisk SuddenObstacleRisk

CarOppositeDirectionHIT Day Foggy BehaviorAnomaly	ReducedVisibilityRisk LaneIntrusionRisk CollisionRisk	ReducedVisibilityRisk CollisionRisk LaneIntrusionRisk OcclusionRisk	CollisionRisk ReducedVisibilityRisk
BusObscuresCar DayClear BehaviorAnomaly	OcclusionRisk UnpredictableBehaviorRisk LaneIntrusionRisk CollisionRisk	CollisionRisk OcclusionRisk LaneIntrusionRisk PedestrianRisk	CollisionRisk LaneIntrusionRisk OcclusionRisk UnpredictableBehaviorRisk

Table 3.1.: Gemini Risk Comparison

ChatGPT			
Scenario Name	Ground Truth Risk	Baseline Output	Ontology Grounded Output
StaticHighwayAd4 NightRainy StateAnomaly	ReducedVisibilityRisk LowFrictionRisk LaneIntrusionRisk TrafficViolationRisk	ReducedVisibilityRisk LowFrictionRisk TrafficViolationRisk LaneAmbiguityRisk PerceptionErrorRisk	ReducedVisibilityRisk ObstructionRisk TrafficViolationRisk CollisionRisk
FallenTree DayClear EvidenceBasedAnomaly	ObstructionRisk OcclusionRisk TrafficViolationRisk CollisionRisk	ObstructionRisk OcclusionRisk CollisionRisk TrafficInteractionRisk	OcclusionRisk CollisionRisk ObstructionRisk
FootballFollowedByBoy DayFoggy EvidenceBasedAnomaly	ReducedVisibilityRisk OcclusionRisk SuddenObstacleRisk PedestrianCollisionRisk UnpredictableBehaviorRisk	ReducedVisibilityRisk OcclusionRisk SuddenObstacleRisk PedestrianCollisionRisk CollisionRisk	CollisionRisk ReducedVisibilityRisk OcclusionRisk SuddenObstacleRisk PedestrianCollisionRisk
CarOppositeDirectionHIT Day Foggy BehaviorAnomaly	ReducedVisibilityRisk LaneIntrusionRisk CollisionRisk	ReducedVisibilityRisk CollisionRisk LaneDeviationRisk OcclusionRisk	OcclusionRisk ReducedVisibilityRisk CollisionRisk
BusObscuresCar DayClear BehaviorAnomaly	OcclusionRisk UnpredictableBehaviorRisk LaneIntrusionRisk CollisionRisk	CollisionRisk OcclusionRisk LaneIntrusionRisk TrafficInteractionRisk	CollisionRisk LaneIntrusionRisk ObstructionRisk

Table 3.2.: ChatGPT Risk Comparison

LLaMA			
Scenario Name	Ground Truth Risk	Baseline Output	Ontology Grounded Output
StaticHighwayAd4 NightRainy StateAnomaly	ReducedVisibilityRisk LowFrictionRisk LaneIntrusionRisk TrafficViolationRisk	ReducedVisibilityRisk LowFrictionRisk CollisionRisk PerceptionErrorRisk	ReducedVisibilityRisk SuddenObstacleRisk CollisionRisk
FallenTree DayClear EvidenceBasedAnomaly	ObstructionRisk OcclusionRisk TrafficViolationRisk CollisionRisk	CollisionRisk OcclusionRisk TrafficComplexityRisk	CollisionRisk LaneIntrusionRisk
FootballFollowedByBoy DayFoggy EvidenceBasedAnomaly	ReducedVisibilityRisk OcclusionRisk SuddenObstacleRisk PedestrianCollisionRisk UnpredictableBehaviorRisk	ReducedVisibilityRisk CollisionRisk SuddenObstacleRisk PedestrianCollisionRisk	ReducedVisibilityRisk CollisionRisk SuddenOstacleRisk
CarOppositeDirectionHIT DayFoggy BehaviorAnomaly	ReducedVisibilityRisk LaneIntrusionRisk CollisionRisk	ReducedVisibilityRisk CollisionRisk ReactionDelayRisk	ReducedVisibilityRisk CollisionRisk
BusObscuresCar DayClear BehaviorAnomaly	OcclusionRisk UnpredictableBehaviorRisk LaneIntrusionRisk CollisionRisk	CollisionRisk OcclusionRisk LaneIntrusionRisk MultiVehicleInteraction Risk	CollisionRisk LaneIntrusionRisk OcclusionRisk UnpredictableBehaviorRisk

Table 3.3.: LLaMA Risk Comparison

Table 3 presents a qualitative comparison between ground truth risks, baseline outputs, and ontology-grounded outputs across different scenarios. The baseline models generally capture most relevant risks but often include additional risks not present in the ground truth, leading to inconsistencies and potential hallucination. In contrast, the ontology-grounded outputs are more aligned with predefined risk classes, resulting in more consistent and structured predictions.

While baseline outputs are more flexible, they are also more prone to inconsistency and hallucination. The ontology-based approach, although slightly more constrained, produces more reliable, interpretable, and semantically consistent results.

Overall, the comparison highlights a trade-off between flexibility and reliability, with ontology-grounded models offering improved consistency and reduced hallucination.

Model	Batch	Baseline Precision	Precision	Baseline Recall	Recall	Baseline F1	F1	Baseline hallucination Rate	Hallucinated Risk Rate
ChatGPT-5.4 mini	1	0.86	0.80	0.81	0.38	0.83	0.51	0.14	0.20
ChatGPT-5.4 mini	3	0.83	0.84	0.81	0.41	0.81	0.54	0.17	0.16
ChatGPT-5.4 mini	5	0.81	0.92	0.68	0.48	0.74	0.63	0.19	0.08
Gemini 3 Flash Preview	1	0.84	0.93	0.74	0.58	0.76	0.71	0.11	0.07
Gemini 3 Flash Preview	3	0.87	0.98	0.79	0.62	0.83	0.75	0.1	0.02
Gemini 3 Flash Preview	5	0.89	0.97	0.81	0.70	0.84	0.81	0.09	0.03
LLaMA	1	0.48	0.57	0.71	0.30	0.57	0.20	0.52	0.22
LLaMA	3	0.55	0.40	0.68	0.21	0.61	0.27	0.45	0.23
LLaMA	5	0.6	0.50	0.66	0.25	0.63	0.33	0.4	0.50

Table 1: Performance Of Different Models Across Varying Batch Sizes

This table compares the performance of baseline and ontology-grounded models across different batch sizes using precision, recall, F1 score, and hallucination rate. The results show that baseline models generally achieve higher recall and F1 scores, indicating that they detect a larger number of risks. However, this comes at the cost of higher hallucination rates.

In contrast, the ontology-grounded models demonstrate lower hallucination rates across all models, particularly for Gemini 3 Flash Preview, where hallucination is significantly reduced. This indicates that the ontology helps constrain the model to generate more reliable and semantically valid outputs. While there is a slight reduction in recall and F1 score in some cases, the outputs are more consistent and aligned with the predefined ontology structure.

Overall, Baseline models are more flexible and detect more risks, whereas ontology-grounded models produce more accurate, structured, and trustworthy predictions, making them more suitable for safety-critical applications.

Model	Scenario	Weather Condition	baseline F1	F1 Score	baseline hallucination	Hallucination Rate
Gemini 3 Flash Preview	StateAnomaly StaticHighwayAd4	Night Rainy	0.87	0.79	0.09	0.04
Gemini 3 Flash Preview	EvidenceBasedAnomaly FallenTree	Day Clear	0.81	0.69	0.11	0.11
Gemini 3 Flash Preview	EvidenceBasedAnomaly FootballFollowedByBoy	Day Foggy	0.84	0.65	0.09	0.07
Gemini 3 Flash Preview	BehaviourAnomaly CarOppositeDirectionHIT	Day Foggy	0.78	0.83	0.11	0.00
Gemini 3 Flash Preview	BehaviourAnomaly BusObscuresCar	Day Clear	0.74	0.71	0.1	0.02
ChatGPT 5.4 mini	StateAnomaly StaticHighwayAd4	Night Rainy	0.79	0.48	0.13	0.24
ChatGPT 5.4 mini	EvidenceBasedAnomaly FallenTree	Day Clear	0.69	0.57	0.44	0.17
ChatGPT 5.4 mini	EvidenceBasedAnomaly FootballFollowedByBoy	Day Foggy	0.67	0.48	0.42	0.13
ChatGPT 5.4 mini	BehaviourAnomaly CarOppositeDirectionHIT	Day Foggy	0.64	0.68	0.36	0.11
ChatGPT 5.4 mini	BehaviourAnomaly BusObscuresCar	Day Clear	0.75	0.45	0.18	0.22
LLaMA	StateAnomaly StaticHighwayAd4	Night Rainy	0.75	0.39	0.18	0.27
LLaMA	EvidenceBasedAnomaly FallenTree	Day Clear	0.42	0.23	0.58	0.41
LLaMA	EvidenceBasedAnomaly FootballFollowedByBoy	Day Foggy	0.60	0.40	0.33	0.27
LLaMA	BehaviourAnomaly CarOppositeDirectionHIT	Day Foggy	0.62	0.48	0.32	0.19
LLaMA	BehaviourAnomaly BusObscuresCar	Day Clear	0.59	0.26	0.34	0.13

Table 2: Risk Discovery

This table presents a scenario level comparison of baseline and ontology grounded performance across different models, using F1 score and hallucination rate as key metrics. For Gemini 3 Flash Preview, the ontology-based approach generally maintains competitive F1 scores while

significantly reducing hallucination. For example, in the StaticHighwayAd4 scenario, hallucination decreases from 0.09 to 0.04, and in CarOppositeDirectionHIT, it is reduced to 0.00 while even improving F1 score from 0.78 to 0.83. This indicates that ontology grounding enhances reliability without major loss in accuracy for this model.

For ChatGPT 5.4 mini, the ontology-based approach leads to a noticeable reduction in hallucination across most scenarios (e.g., 0.44 to 0.17 in FallenTree), but this is accompanied by a decrease in F1 score in several cases. This suggests that while the ontology improves consistency, it also constrains the model, limiting its ability to identify a broader range of risks.

Similarly, for LLaMA, the ontology-based approach reduces hallucination (e.g., 0.58 to 0.41 in FallenTree), but also results in lower F1 scores across all scenarios. This indicates improved reliability but reduced overall detection capability.

Overall, the results demonstrate that ontology grounding consistently reduces hallucination across all models but introduces a trade-off with F1 score. The effect is most balanced in Gemini, where reliability improves with minimal loss in accuracy, while for ChatGPT and LLaMA, the hallucination comes at a more noticeable cost to performance.

Limitations

While the ontology-grounded approach improves semantic consistency and reduces hallucination, it also has some limitations.

First, the ontology introduces a constraint on the model's flexibility. Because the model is guided to use predefined classes and relationships, it may fail to identify valid risks that are not explicitly represented in the ontology. This can lead to a reduction in recall and F1 score, particularly in scenarios with complex or uncommon hazards.

Second, the performance of the ontology-grounded approach is highly dependent on the coverage and quality of the ontology. If certain risk types or environmental conditions are missing or not well-defined, the model may be forced to map them to less appropriate classes, reducing semantic accuracy.

Third, the evaluation relies on a manually defined ground truth, which may not capture all valid risks present in a scenario. As a result, some risks identified by the model may be incorrectly counted as false positives, affecting precision and hallucination metrics.

Additionally, the quality of RDF outputs depends on model compliance with the required structure. In cases where the model generates incomplete or syntactically incorrect Turtle output, fallback mechanisms (e.g., regex-based extraction) are required, which reduces the semantic reliability of the evaluation.

Finally, differences across models indicate that not all LLMs respond equally well to ontology constraints. While models like Gemini maintain a balance between accuracy and reliability, others such as LLaMA show a more significant drop in performance when constrained.

Conclusions

This project evaluated the effectiveness of large language models in identifying safety risks in autonomous driving corner-case scenarios, with a focus on comparing baseline and ontology-grounded approaches. The results show that while baseline models are more flexible and often achieve higher recall or F1 scores, they are also more prone to inconsistency and hallucination.

In contrast, the ontology-based approach significantly improves semantic consistency and reduces hallucination across all models. By constraining the model to predefined classes and relationships, the generated outputs become more structured, interpretable, and aligned with domain knowledge. This improvement is particularly evident in models such as Gemini 3 Flash Preview, where hallucination is reduced with minimal impact on overall performance.

However, a trade-off is observed. The ontology-based approach can limit the model's flexibility, leading to a slight reduction in F1 score in some scenarios. Additionally, the evaluation is influenced by the size and coverage of the ground truth, which may not capture all valid risks identified by the models.

Overall, the findings suggest that ontology grounding is a valuable approach for improving reliability in safety-critical applications. While it may constrain the range of detected risks, it enhances consistency, reduces unsupported predictions, and produces outputs that are better suited for structured evaluation and real-world deployment.